

CSE326: Internet & Web Technologies

100-Percentile Master Study Guide, Textbooks Deep-Dive & Curated Video Hub

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1. Standard Reference Textbooks & Deep-Dive Chapter Context

Unit	Subject Area	Standard Textbook	Core Chapter Concept & Key Exam Takeaways
Unit I & II	HTML5 & Forms	Laura Lemay et al. (Mastering HTML, CSS & JS)	Ch 1 & 2: Client-Server HTTP cycle, DOCTYPE, meta viewport, semantic elements (<header>, <nav>, <main>, <article>, <section>, <aside>, <footer>), vector SVG. Ch 5 & 6: Form inputs (email, number, range, date), regex validation (pattern, required), ARIA accessibility.
Unit III	CSS3 Layouts & Animations	HTML5 Black Book (Wiley / DT Editorial)	Ch 7, 8 & 9: Specificity formula (a,b,c,d), Combinators, Pseudo-classes (:hover, :focus, :nth-child()), Pseudo-elements (::before, ::after). Ch 10 & 11: Box Model (content-box vs border-box math), Positioning (relative, absolute, fixed, sticky), 1D Flexbox vs 2D CSS Grid (repeat, minmax, auto-fit), CSS Variables.
Unit IV & V	JavaScript & DOM APIs	Eloquent JavaScript (Marius Haverbeke) / MDN	Ch 2, 3 & 4: Execution Context, Call Stack, Scope (var vs let/const block scope), Hoisting, Closures, Higher-Order functions (map, filter, reduce), Object destructuring. Ch 14, 15 & 18: DOM Tree, Event Bubbling vs Capturing, Event Delegation, Web Storage (localStorage vs sessionStorage), Fetch API async/await.
Unit VI	Deployment & Performance	Pro Git (Scott Chacon) & Chrome DevTools	Chrome DevTools debugging, Network waterfalls, Lighthouse performance audits, Git repository management, Automated static website deployment to GitHub Pages.

2. Curated YouTube Video Playlists & Online Resource Links

Unit / Topic	Recommended Channel	Search Query / Video Reference	Core Takeaway & Key Concepts
Unit I & II: HTML5 & Forms	Dave Gray	Dave Gray HTML Full Course for Beginners	Semantic tags, accessible form controls, regex pattern.
Unit III: CSS Specificity & Box Model	Kevin Powell	Kevin Powell CSS Specificity and Box Model	Specificity formula (a,b,c,d), border-box sizing reset.
Unit III: Flexbox Masterclass	Kevin Powell	Kevin Powell Flexbox CSS Guide	Main/Cross axis, justify-content, responsive navbars.
Unit III: CSS Grid in 20 Minutes	Web Dev Simplified	Web Dev Simplified CSS Grid Tutorial	2D layout, repeat(auto-fit, minmax(280px, 1fr)).
Unit IV: JS Execution Context	Akshay Saini	Namaste JavaScript Akshay Saini Season 1	Execution context, call stack, hoisting, closures.
Unit IV: map, filter, reduce	Akshay Saini	Akshay Saini map filter reduce JavaScript	Array transformations and functional accumulation.
Unit V: DOM & Event Delegation	Traversy Media	Traversy Media JavaScript DOM Crash Course	querySelector, Event Bubbling, Event Delegation.
Unit V: Fetch API & Async/Await	Web Dev Simplified	Web Dev Simplified Fetch API JavaScript	HTTP GET/POST, async/await, try/catch error handling.
Unit VI: GitHub Pages Deployment	Traversy Media	Deploy Website to GitHub Pages Traversy Media	Git CLI commit, push, and free static hosting.

3. High-Yield Unit Summaries, Diagrams & Formulas

Unit I & II (HTML5 & Forms): Client-Server HTTP request-response cycle and Critical Rendering Path (DOM + CSSOM → Render Tree → Layout → Paint). HTML5 Semantic structure (<header>, <nav>, <main>, <article>, <section>, <aside>, <footer>). Forms use strict native validation via pattern (Regex), required, and semantic input controls (email, number, range, date).

Unit III (CSS3 Mastery): Specificity tuple (a,b,c,d) = Inline (1000) > ID (100) > Class/Attribute/Pseudo-class (10) > Element/Pseudo-element (1). Box model: box-sizing: border-box absorbs padding and borders inside width. 1D Flexbox (Main/Cross axis, justify-content, align-items) vs 2D CSS Grid (grid-template-columns: repeat(auto-fit, minmax(280px, 1fr))).

Unit IV (JavaScript Core): Single-threaded, synchronous execution context (Memory Creation Phase with Hoisting → Code Execution Phase). var is function-scoped; let/const are block-scoped and held in Temporal Dead Zone (TDZ). Higher-order methods: .map(), .filter(), .reduce().

Unit V (Interactive DOM & Web APIs): DOM Event Propagation follows Capturing (top-down) → Target → Bubbling (bottom-up). Event Delegation attaches a single listener to a common ancestor using event.target.matches(). Web Storage: localStorage (5MB, persists forever) vs sessionStorage (5MB, tab session) vs Cookies (4KB, sent in HTTP headers). Fetch API with async/await and try/catch.

Unit VI (Deployment & Best Practices): Debugging with Chrome DevTools (Console, Breakpoints, Network waterfalls). Static web hosting via GitHub Pages (git push origin main → automated live deployment at username.github.io/repo).

4. 100-Percentile Exam Presentation Blueprint

Tier 1: Formal Technical Definition (1-2 crisp lines with core web terminology bolded)
Tier 2: Labeled Architectural Diagram / DOM Tree (Neat boxed ASCII / flow diagram)
Tier 3: Comparative Matrix / Specificity Calculation Tuple (Highlighting parameters)
Tier 4: Production-Grade Code Snippet (HTML5/CSS3/ES6 code with error handling)